

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that phase order alone establishes neither shared content nor a self-model, and state what additionally does. We separate three conditions: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared Lipschitz encoders, and the bound grows as the square root of the population read: at the patch order of a simulated cortical sheet it constrains at most 10 sites, so on a larger population the positive link from coherence to content rests entirely on a bridge assumption. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor; a broad family needs input-conditioned maps

and discriminating codes. Lean 4 verifies these implications, and one model satisfies all eight connecting hypotheses together; this shows consistency, not cortical realization. The conditions yield a disconfirmable comparison: the agreement of decoders fitted independently on two regions, in excess of what their separate accuracies imply, should be larger for perceived than for unperceived stimuli at matched phase coherence and single-region accuracy, with perception classified by a no-report marker. An exploratory EEG analysis is a measurement check only; its pooled estimator places a perfectly synchronized signal at zero coherence. Resource and implementation constraints, and a held-out onset comparison during anaesthetic emergence, are stated briefly. No principle yet selects the cover, and the sheaf’s overlaps are spatial where the example’s are overlaps of content, so the proposed correlate of unity has no operational definition. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015) and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction

13 from an internal encoding). The mathematical states are spatial measures
14 and fields; identifying them with perceptual content requires independent
15 decoding evidence.

16 Binding-by-synchrony and communication-through-coherence accounts
17 treat phase relations as the medium of integration or of selective routing
18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
20 leave open what fact about synchronized activity makes it one content rather
21 than coordinated activity carrying several. The contribution here is to sep-
22 arate the three requirements and prove which follow from which. Phase
23 order constrains shared content only through encoders related on overlaps,
24 with a bound that grows as the square root of the population it is read
25 on. Compatible local descriptions, not synchrony, determine a unique global
26 state. A contracting self-map fixes one state but can reconstruct accurately
27 only a family of bounded diameter. Each result names a measurement that
28 phase data alone cannot supply.

29 The objection that synchrony does not fix content is older than this work.
30 Critiques of the temporal binding hypothesis argued that synchrony, where
31 it occurs, cannot say which features belong together or what they jointly
32 represent: that grouping has to be computed before synchrony could sig-
33 nal it (Shadlen & Movshon, 1999). The formal results here are elementary:
34 uniqueness of gluing, Banach’s fixed-point theorem, the triangle inequality
35 and a pigeonhole count. What formalization adds to the informal argument is

36 quantitative and bookkeeping content. It gives the rate at which a coherence
37 bound on content weakens with population size and the population at which
38 it stops constraining anything, the diameter within which one contracting
39 map can reconstruct, and the number of codes a separated family of states
40 requires. It also gives a complete inventory of the premises a synchrony-to-
41 content argument consumes, each of which names a measurement, together
42 with one model in which all of them hold at once.

43 The familiar noisy mean-field oscillator supplies a stationary coherence
44 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991). We add a resource
45 condition for a declared actuator, bounds that connect phase order to con-
46 tent only through specified encoders, and conditions under which compatible
47 descriptions glue and an accurate reconstruction fixes the glued state. A
48 Lean 4 development checks these implications and a joint witness satisfies
49 their eight connecting hypotheses in a specified model. It proves neither that
50 cortex satisfies those hypotheses nor that the glued state is experienced. The
51 latter identification motivates the framework and needs its own endpoint.

52 The empirical question is which measurements could support or discon-
53 firm the proposed connections. A phase resultant can miss a locally or-
54 dered winding field; stationary phase distributions can leave the coupling
55 mechanism unidentified; and coherent adaptation can lose alignment with
56 a designated content template. Independent decoders must identify shared
57 quantities before their agreement can be tested. We therefore distinguish sta-
58 tionary branches from finite observed onset, and compare prespecified field

59 and connectivity onset models on held-out emergence recordings. The com-
60 parison tests a specified spatial prediction, not the presence of experience.
61 Section 10.2 states the comparison that bears on conscious content directly.
62 Sections 2–5 develop the three requirements; Section 6 applies their mea-
63 surement obligations to a digital candidate. Section 7 states what the formal
64 composition establishes, and Sections 8–9 examine observables and the cor-
65 tical test. Table 1 lists the connecting hypotheses the article argues from;
Supplemental Material gives the full chain, derivations and controls.

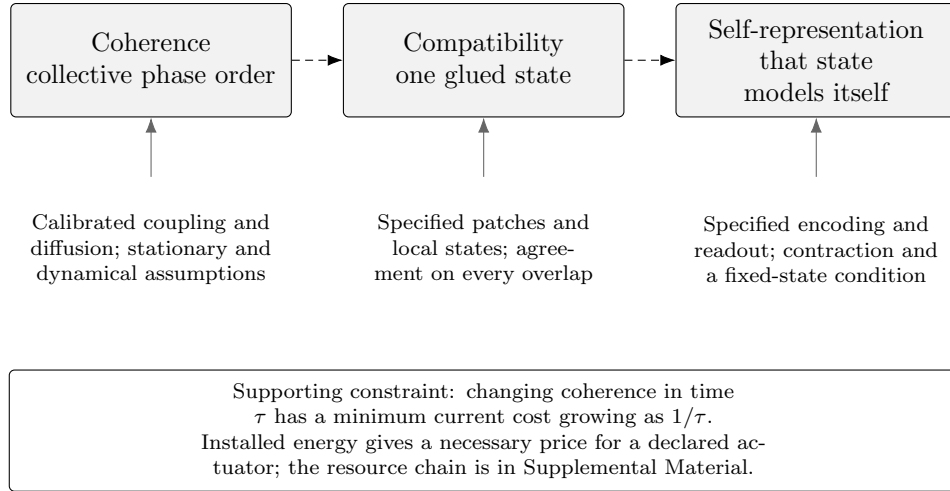


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the connecting hypotheses of the formal composition that the article argues from.

67 **2 Limits of phase order**

68 Several limits hold without the cortical hypothesis. Equal phases can coexist
69 with incompatible local descriptions; phase order constrains shared content
70 only through encoders related on overlaps. A winding field can have zero
71 global resultant while remaining locally ordered. A constant readout can
72 have a unique fixed point without reconstructing its input. A region that
73 does not distinguish relevant states cannot encode them, and a site cannot
74 report a change outside its causal past by a declared deadline. These lim-
75 its reflect familiar phase oscillator and information constraints (Acebrón et
76 al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017).
77 The sections below state each one beside the condition or measurement it
78 motivates.

79 **3 Coherence: physical coupling and its cost**

80 **3.1 Dynamics and the stationary threshold**

81 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

82 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
83 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,

84 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
 85 quency heterogeneity and diffusion are distinct parameters. The stationary
 86 calculation below uses identical frequencies and this scalar coupling. Ap-
 87 plying it to a spatial kernel requires an aggregation rule and a check of the
 88 resulting approximation.

89 For identical frequencies in their rotating frame, the positive normalized
 90 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

91 where I_n are modified Bessel functions and ψ is the collective phase. The for-
 92 mal derivation uses the absence of stationary current for a periodic gradient
 93 drift to obtain this density, then verifies normalization and self-consistency.
 94 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz &
 95 Mirollo, 1991). Above threshold, exactly one positive solution coexists with
 96 zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

97 Figure 2 shows this stationary branch.

98 Linearization about the uniform density gives rate $K/2 - D$ for the first
 99 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 100 exactly above threshold. About the coherent density, the quadratic form
 101 is strictly dissipative transverse to collective rotation, whose rate is zero.

102 These linear estimates concern the limiting equation. Relating them to finite
 103 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 104 tion; exchange symmetry alone does not establish the needed independence
 105 (Sznitman, 1991).

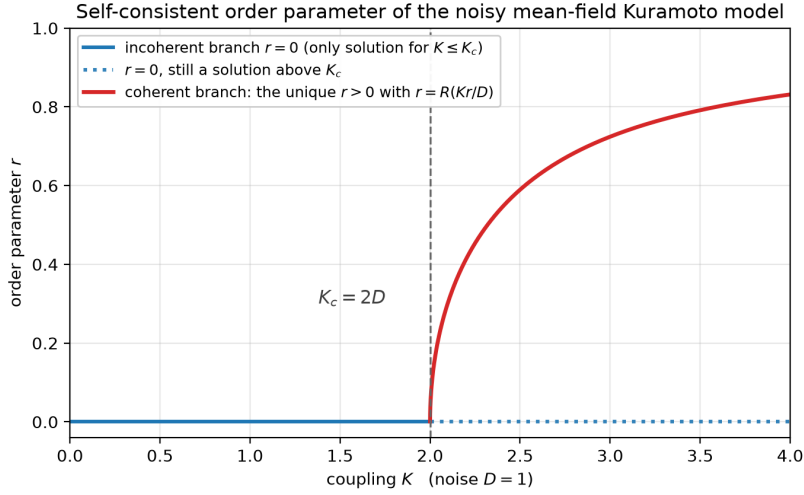


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

106 3.2 A speed limit on changes of coherence

107 One thermodynamic statement bears on coherence directly. For a smooth
 108 positive periodic phase density with probability current J and diffusion D ,

109 the current functional $\sigma_J = \int J^2/(D\rho) d\theta$ satisfies

$$\int_0^\tau \sigma_J dt \geq \frac{[\arcsin r(\tau) - \arcsin r(0)]^2}{D\tau} \quad (4)$$

110 while $0 < r < 1$. A change of coherence completed in time τ therefore has a
111 minimum current cost that grows as $1/\tau$, whereas holding a stationary order
112 has no positive floor. Read the other way, a current budget C sets a minimum
113 duration $\tau \geq [\Delta \arcsin r]^2/(DC)$ for a given change. This bears on ignition-
114 like transitions, in which large-scale coherence rises abruptly (Dehaene et al.,
115 2014), and on recovery latency after anaesthesia: each has a floor set by the
116 current available to drive it. The bound is a heat or power statement only
117 when J is identified with a physical flux, a reservoir at a stated temperature
118 is specified and local detailed balance holds for the steps concerned; only
119 then does σ_J convert to entropy production (Landauer, 1961; Norton, 2005;
120 Shenker, 2000). Installed energy supplies a separate, necessary price: for
121 a declared actuator whose response modes carry installation prices, mean
122 coupling is at most κU_{inst} for a mode-data factor κ , so a coherent stationary
123 branch requires installed energy $U_{\text{inst}} > 2D/\kappa$, and no cortical value of κ
124 is known. Supplemental Material gives both derivations, the register and
125 feedback budgets, and the sensitivity of the metabolic comparison to its
126 assumptions.

127 4 Compatibility: from coordinated activity 128 to a global description

129 4.1 Why common timing is insufficient

130 For overlapping patches U and V , compatibility means that restricting either
131 local description to $U \cap V$ gives the same object. Descriptions can agree on
132 shared quantities while contributing different information elsewhere. Com-
133 patible sections of the specified sheaf have a unique global extension. On
134 a finite cover, local finite spatial measures agreeing as measures on every
135 overlap also have a unique finite-measure extension whose sheaf image is the
136 glued section. No mass-one normalization is imposed.

137 In the cup example, visual and reaching descriptions must express a
138 shared quantity, such as cup location, in common coordinates. Restriction
139 retains that quantity while dropping details particular to each description.
140 Agreement permits surface appearance and hand configuration to contribute
141 to one situation. The mathematical counterexample assigns the same phase
142 to every patch while two local mass profiles disagree on a shared site. Those
143 profiles have no common extension. Because they are chosen independently
144 of phase, synchrony places no constraint on their disagreement.

145 This distinction limits the biological interpretation. Receptive territories
146 and content domains must be related to spatial patches, and the descriptions
147 must support the required extension. Matching overlap marginals of de-

148 coded probability laws need not determine a unique global joint distribution
149 (Abramsky & Brandenburger, 2011).

150 The cover is itself a choice, and the construction constrains it in one way
151 only: synchronized patches agree where they overlap. Where they do not
152 overlap there is nothing to compare, so a cover by pairwise disjoint patches
153 satisfies that obligation for *every* assignment of local descriptions and phases
154 and the gluing theorem still returns a unique global section, which is the
155 tuple it was handed. Membership of the class then reports nothing about the
156 system beyond the patches covering it. What a gluing claim asks of a cover
157 is therefore a non-degeneracy the theorem does not assume: that distinct
158 patches meet, and that what they share is a quantity fixed independently of
159 the agreement it is being used to establish.

160 The cup example exposes a second gap, between two kinds of overlap.
161 The sheaf is built over a spatial substrate, so its patches overlap where they
162 share sites. Visual and motor territories that agree on cup location overlap
163 in what they describe, not in where they lie, and the construction supplies
164 no relation between the two. One route recasts the construction over content
165 variables: each region becomes a partial decoder whose domain is the set of
166 variables it estimates, and two regions overlap exactly where they estimate
167 a common variable. The gluing argument needs only a cover and restriction
168 maps, so it applies formally to such a cover, but no result here is stated
169 for one, and which variables and decoders form it is an empirical question.
170 Until a principle selects the cover, the proposed correlate of unity has no

171 operational definition: gluing on a declared cover is proved, and which cover
 172 cortex realizes is open.

173 4.2 When coherence does constrain content

174 A declared relation between phase and content gives a positive result. Let
 175 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 176 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 177 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 178 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 179 sites; their equality on overlaps is a supplied relation between descriptions of
 180 the shared quantity. Write $z_k = e^{i\theta_k}$ and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The
 181 phasors' squared deviations from their mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$,
 182 and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$ gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair.
 183 Consequently every overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1 - r^2)}. \quad (5)$$

184 Here discrepancy is the norm of the difference, and r and N are read on
 185 whichever population the bound is applied to. That choice is what decides
 186 whether the estimate says anything. A patch carries an order parameter
 187 of its own, and reading the bound there carries that patch's resultant and
 188 that patch's site count, so a large population observed through small patches
 189 is constrained by the small count. Read instead on the whole array, the

190 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
 191 from an average over N sites. The order cannot be improved in general:
 192 one oscillator in antiphase among N leaves $N(1 - r^2)$ near 4. At $r = 1$ the
 193 observations agree exactly at either scale. A shared scalar encoder is a special
 194 case. Dropping the overlap relation permits disagreement even at perfect
 195 locking. A common nonexpansive linear projection preserves agreement, but
 196 agreement of a projection does not establish agreement of the full vector.

197 What Eq. (5) bounds is a chord, which never exceeds 2, so the estimate
 198 constrains anything only where the right-hand side falls below that. At the
 199 sheet's own patch order, the bound is informative for at most 10 sites; beyond
 200 that, the coherence-to-content link requires the bridge assumption E78 of
 201 the cortical hypothesis (Table 1 gives the full composition and its edges).
 202 Supplemental Material provides the detailed derivation of this saturation,
 203 and states three conditional derivations that replace the uniform factor $\sqrt{N}$
 204 by a count of disagreeing pairs, a chain across a cover's shared sites, or a
 205 spectral gap at a frequency-locked configuration.

206 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 207 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 208 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 209 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (6)$$

210 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-

211 duce an initial mismatch. Approximate observations leave a nondecaying
 212 term in the error bound. Taking a uniform bound on Eq. (5) connects this
 213 update to the phase process.

214 4.3 What approximate agreement determines

215 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 216 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 217 Weights that vanish outside their patches and sum to one at each site select
 218 a global profile by weighted averaging. The selection lies within ε of each
 219 patch there, and any two such selections differ by at most ε , independently
 220 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 221 $L\varepsilon$ across selections.

222 At zero tolerance, the selection is the unique exact extension. At every
 223 positive tolerance there are families with no exact extension; selection then
 224 gives an approximation whose dependence on weights is bounded. Supple-
 225 mental Material gives a three-site witness attaining the bound. The mass-
 226 profile construction uses the scalar case of the content dynamics and addi-
 227 tionally requires nonnegativity of the initial profiles and observations. The
 228 general vector result supplies no identification with the spatial-measure sheaf.

229 These constructions concern declared objects: a specified cover, contents
 230 valued in a normed space, and encoders related on overlaps. They select
 231 neither the neural cover nor the decoder, and identifying their contents with
 232 perceptual content, or establishing that the identification bears on experi-

233 ence, is a further modelling task; Section 8 sets out what a measurement of
 234 it would have to establish.

235 **5 Self-representation: a state that models it-** 236 **self**

237 **5.1 The reflexive construction**

238 Compatibility gives a global state. Reflexivity relates it to an encoding within
 239 that state. Equip global sections with a complete metric. If the additional
 240 restriction-resonance condition identifies encoding with restriction to a region
 241 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (7)$$

242 The region and readout are fixed across the state space. If F contracts and
 243 fixes the section s_* formed from the local descriptions, Banach's theorem
 244 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
 245 (Table 1 gives the full composition and its edges) supplies the contraction
 246 and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling
 247 makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for
 248 a relaxation time $\tau > 0$. The factor is stipulated, not derived. It borrows
 249 the rate $(K - 2D)/2$ at which the first harmonic of the uniform density

grows (Section 3), a rate about a different state from the one F acts on,
and no result derives a Lipschitz law for an encoding or readout from the
phase dynamics. Constructing a map with that factor remains a modelling
obligation. A fixed contraction can reconstruct accurately only a restricted
family of states: if its error is at most ε throughout that family, any two
members are at most $2\varepsilon/(1 - \Lambda)$ apart.

In the cup example, the global state includes the represented scene and the
perceiver’s relation to it. A local encoding must retain enough information
for the readout to recover that modelled state; cup position alone would be
insufficient if hand configuration and viewpoint could not be reconstructed.

For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruction gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (8)$$

At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for
fixed positive ε and τ . Exact uniqueness survives while the error guarantee
weakens.

5.2 Reconstruction across a family of states

Sensitivity can be required quantitatively. Declare relevant macrostates, an
encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that

268 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (9)$$

269 Thus M relevant states pairwise more than 2ε apart require at least M
 270 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 271 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 272 existing self-prediction map.

273 This is an additional accuracy criterion on a family, with no requirement
 274 for multiple exact fixed points. For the same fixed contracting map used
 275 by E89, however, the diameter bound above is an additional restriction: at
 276 zero error the family contains at most one metric-distinct state. A broad
 277 family therefore requires a different, input-conditioned reconstruction map,
 278 one map F_u with its own fixed point for each input u , with its own accuracy
 279 and stability assumptions. No such map is formalized or constructed here.
 280 Under E89 the proposed minimal-self correlate is therefore a condition on
 281 one self-state, not a model of a self that follows a changing scene. A blind
 282 encoder or constant readout fails the stated criterion for separated states
 283 despite having a unique fixed point.

284 Where the encoding is read as the state's own restriction to a region, the
 285 criterion becomes a condition on the region. Restriction resonance requires
 286 the encoder and that restriction to identify the same pairs of states, so a
 287 boundary reconstructing two relevant states more than 2ε apart while its

288 region reads them identically is not resonant. Read forwards, a resonant
 289 boundary that reconstructs a pairwise separated family restricts injectively
 290 on it. Which states a region reads identically is fixed by where the region
 291 sits, before an encoding is chosen, so the resolution the family demands is
 292 the region's and no readout supplies it. The implication runs one way: a
 293 region that does resolve the family is not thereby shown to carry a resonant
 294 encoding, and E89 supplies no such resolution.

295 Resonance and resolution are jointly satisfiable without a local fold. Take
 296 the region to be the whole substrate and use the identity readout: restric-
 297 tion is the identity, so this boundary is resonant and reconstructs every de-
 298 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 299 E89's strict contraction premise on a nontrivial metric space. A properness
 300 condition on the region is therefore a separate requirement, and the fixed-
 301 contraction diameter bound remains in force when E89 is also required. With
 302 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 303 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 304 so the family is no larger than the δ -packing number of what the region car-
 305 ries. That is the shape a continuous family of states needs, and at the whole
 306 substrate it says nothing, because the region's range is everything. The gain
 307 L is declared and derived here for no physical readout, and at unbounded
 308 gain the bound says nothing.

309 **5.3 Agreement after a change takes communication**

310 Snapshot compatibility does not specify how quickly descriptions respond
311 when a shared quantity changes. In a synchronous network whose updates
312 read only the site’s own state and declared neighbours’ messages, two initial
313 configurations agreeing on a site’s causal past through T rounds produce the
314 same state there at round T . The deadline consequence is the one Section 2
315 states, and it follows from the reconstruction bound in Section 5.2: a report
316 from that site is a readout of a code that the two interventions leave identical.

317 A three-site delay line witnesses failure after one round and exact report-
318 ing after two; cutting the path prevents delivery at every round. A report
319 may still happen to be correct about one fixed world. The bound concerns
320 responsiveness across interventions.

321 Throughout this section *prediction* names a reconstruction map, and the
322 relevant family of states, the metric, the tolerance, the communication graph
323 and the deadline are all declared inputs, as are the represented variables and
324 the fidelity counted as accurate. Reading the map as temporal neural pre-
325 diction would require dynamics; reading the resulting code count as physical
326 memory capacity would require a hardware model, which a state-dependent
327 readout would have to extend.

328 6 What the conditions constrain in a digital 329 candidate

330 The conditions bear on a large language model (LLM) running on graphics
331 processing units (GPUs) only through measurements, and only for a declared
332 task. For a finite spatial coupling operator with eigenvalues $|\lambda_1| \geq \dots \geq |\lambda_N|$,
333 any linear encoder and readout factoring through d real coordinates pays re-
334 construction error at least $\sum_{i>d} \lambda_i^2$. That rank premise is a property of an
335 explicitly linear bottleneck; query/key dimension does not supply it for soft-
336 max attention. On one synthetic two-view task whose target is the glued
337 state and a self-relative coordinate, a rank-two linear decoder has predicted
338 training-error floor 2.000 and held-out error 2.000 on a parity-held-out family.
339 The deadline premise is similarly specific: within one forward pass, the causal
340 mask confines position t to its causal past, so an intervention outside it can-
341 not be reconstructed there by that step. These results rank no architecture
342 family, imply no general inferiority of GPU hardware, and settle no question
343 of machine experience; a claim of unity or self-representation for such a sys-
344 tem commits its maker to identifying the shared quantities, the code and its
345 separation of the declared family under intervention. Supplemental Material
346 gives the bottleneck derivation, the task controls, the information ceiling of
347 token outputs and the hardware scope.

348 7 The formal composition

349 The Lean development composes the three requirements into one conditional
350 theorem. Its passive and feedback branches each accept eight connecting hy-
351 potheses, running from finite capacity through a register, dissipation and a
352 continuum coupling to the field, coherence, the cover and the self-prediction
353 fixed point. Each edge is supplied rather than derived from the one above it,
354 and the inferential content of the composition is its last step, where contrac-
355 tion and a fixed glued section give uniqueness. Table 1 lists the four edges the
356 article argues from; Supplemental Material gives the full chain, its thermo-
357 dynamic edges and the register ledger of the feedback branch. Both branches
358 also take a cover carrying exact overlap compatibility and a reflexive bound-
359 ary specifying encoding and readout, and conclude that the glued section is
360 the unique fixed point of self-prediction. Composing the edges establishes
361 that the premises are consistent, named in one place and machine-checked
362 together; it does not establish that self-representation follows from coher-
363 ence. Joint witnesses satisfy every edge of each branch in a specified model,
364 and establish no common cortical mechanism. The development declares no
365 additional axioms beyond its libraries' foundational axioms.

Table 1: The four connecting hypotheses the article argues from, shared by both branches of the formal composition. Supplemental Material gives the full eight-edge chain and, in Table S1, maps the node-level results.

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover’s nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve’s diameter; where the configuration is frequency-locked it carries through the coupling graph’s algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.

366 8 What the observations establish

367 The composition holds under its premises. Whether observation can estab-
368 lish those premises is a separate question, and the three requirements it
369 connects concern different objects: phase dynamics, local descriptions and
370 internal encodings. Their observables must retain these distinctions. The
371 following controls show which mechanisms remain indistinguishable, which
372 temporal approximations fail, which states one observable cannot separate,
373 and what independent content measurements must establish. Each is a finite
374 numerical regime on a declared model, with full parameters and controls in
375 Supplemental Material, and none of them calibrates a cortical quantity.

376 8.1 Stationary distributions leave the mechanism un- 377 determined

378 Equation (2) gives a concentration–coherence relation with no fitted shape
379 parameter. Estimate r from circular resultant length and a from density
380 shape: inverting R from the measured r would make agreement automatic.
381 Even independently estimated agreement identifies a statistical family shared
382 by different mechanisms.

383 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
384 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
385 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
386 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

387 and diffusion D_0 have that same density. Their stationary distributions co-
 388 incide window by window despite different causes.

389 Resolved dynamics supply additional information. In the ideal model
 390 with continuous latent paths, quadratic variation accumulates at rate $2D$
 391 and conditional increment means constrain drift; a filtered observation of
 392 the phase needs its own observation model before either is read off. Separat-
 393 ing endogenous coupling from a matched common drive additionally requires
 394 a measured drive or controlled perturbation; identifying electromagnetic cou-
 395 pling requires evidence specific to that interaction. Supplemental Material
 396 gives the equivalence construction and its numerical check.

397 **8.2 Finite ramps change the observable onset**

398 The square-root law in Eq. (3) describes stationary solutions. A changing
 399 coupling requires time for the phase distribution to approach those solutions.
 400 In a deterministic mean-field control that crosses the threshold symmetrically,
 401 none of the scanned rates meets the declared tracking tolerance: slowing
 402 the crossing does not remove the terminal residual under that initialization.
 403 A slow control variable therefore supplies no general guarantee of branch
 404 tracking at onset. Supplemental Material gives the residual, rate range and
 405 admissible rates away from threshold.

406 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
 407 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
 408 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay

409 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
 410 6 uncensored legs supply the delay scaling below; the decade-spaced ones
 411 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
 412 onset exponents differ from those of the exact stationary branch evaluated
 413 on the same grids. At the fastest speed the ensemble mean never reaches the
 414 fit window, so its fitted slope does not estimate stationary onset.

415 Delayed passage through a noisy bifurcation (Berghlund & Gentz, 2002)
 416 makes the apparent exponent depend on ramp speed and fitting window.
 417 The specified deviation-based critical ramp speed is not bracketed by this
 418 sweep. With fluctuating coupling, the relevant stationary comparison also
 419 depends on rate: the mean coupling approximates order in the tested fast
 420 regime, while averaging stationary order over coupling works for the tested
 421 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 422 neither approximation.

423 **The prediction.** Over the 6 uncensored legs a finite-range fit gives $\Delta K \propto$
 424 $v^{0.444}$. The interval from fitting those means is [0.394, 0.494]; resampling
 425 replicas independently within each speed gives a 95% interval [0.398, 0.496],
 426 with fraction 0.019 of 10000 bootstrap fits at or above 1/2. This fraction
 427 describes the resampling distribution under the fitted protocol; it is not a
 428 test of the ideal exponent. A deterministic mean-field run seeded at the
 429 fluctuation floor gives exponent 0.425 with the same estimator and escape
 430 criterion; tightening its criterion to $r \geq 0.05$ gives 0.490. These controls

431 show that the criterion affects the fitted exponent, but do not prove that it
 432 accounts for the entire finite-population shortfall. The larger-population low-
 433 threshold fit has substantial log residuals; matched-speed fits do not converge
 434 toward one-half across the tested sizes and criteria. Some trajectories also
 435 cross the absolute escape level before K_c , so the first postcritical crossing can
 436 yield a zero or grid-limited delay. This finite sweep establishes no common
 437 asymptotic exponent.

438 The ideal delayed-onset calculation motivates a square-root scaling hy-
 439 pothesis; its exponent is generic to suitable bifurcations and does not iden-
 440 tify the coupling medium. A within-subject emergence design can vary the
 441 propofol taper and test how an operational onset changes with that manipu-
 442 lation, but interpreting the result as a coupling-rate law requires a measured
 443 or calibrated relation between taper and $\dot{K}$. The escape criterion must be
 444 set relative to each recording's pre-threshold fluctuation scale. Section 8.1
 445 describes how D could be estimated from resolved phase traces.

446 Additional controls address integration bias, population size and spatial
 447 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 448 $[1.958, 2.009]$ contains 2.

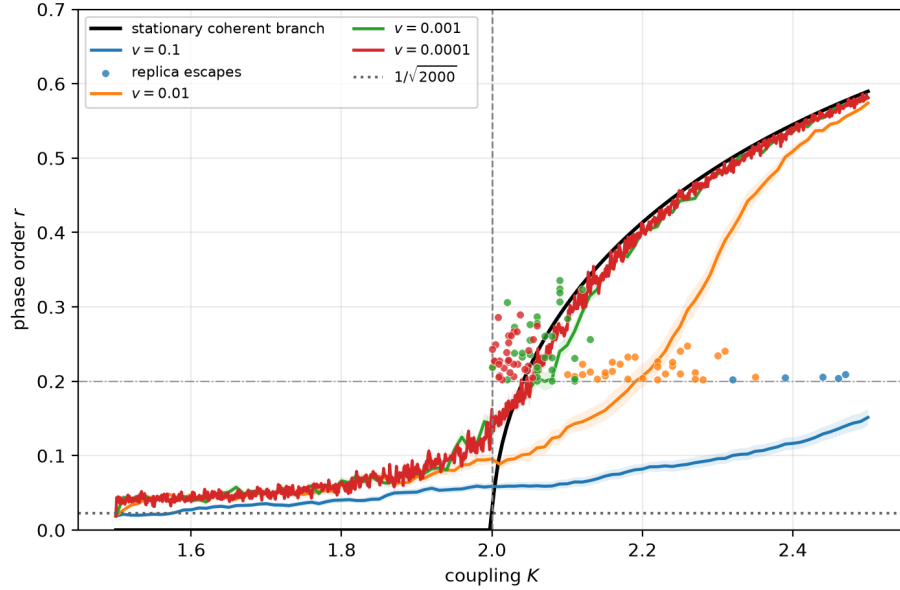


Figure 3: Finite-rate departures from the stationary branch. Curves show ensemble-mean order for four ramp speeds v , with one-standard-error bands. Coloured points mark each replica’s first sustained escape, plotted at its recorded order; censored replicas have no point. The dashed vertical line is the static threshold, the dash-dotted horizontal line is the escape criterion, and the dotted horizontal line is the finite-size reference. The replica-bootstrap interval for the delay exponent is reported in the text.

449 8.3 A winding state leaves the order parameter am- 450 biguous

451 The global order parameter r discards spatial arrangement. On a periodic
452 sheet, a phase field can wind once around the sheet and have $r = 0$ while
453 neighbouring sites remain locked. At identical frequencies, every such wind-
454 ing is stationary for an isotropic symmetric kernel: contributions at opposite
455 separations cancel. It violates the quarter-turn initial-spread premise of the

456 convergence result, which is why that result can still select the uniform state.

457 In the spatial control, the winding persists below the interpolated de-
 458 cay length 0.1871 mm and unwinds above it (Fig. 4). At 0.1 mm, uniform
 459 and winding starts under the same coupling, diffusion and kernel end at
 460 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
 461 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
 462 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
 463 realizations. These are finite controls, not a cortical wave identification or a
 464 derived transition.

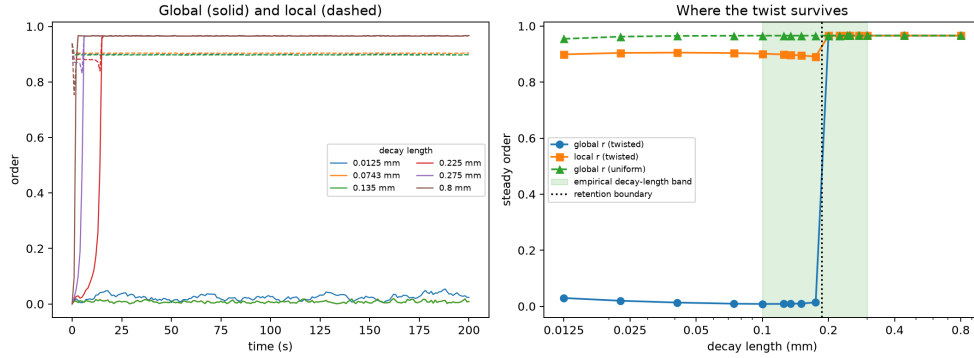


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

465 Linearization about a winding reweights each separation’s coupling by

466 the cosine of its phase advance. The resulting symmetric kernel has explicit
 467 mode rates: nonnegative reweighted coupling gives no growing mode, while
 468 a positive rate destabilizes that winding. The criterion concerns linear rates,
 469 not asymptotic stability, and the measured retention boundary need not
 470 equal its limit. An amplitude extension adds an existence condition: $\Gamma + \lambda >$
 471 0 , where Γ is linear gain and $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ comes from the same
 472 kernel. For the nonnegative kernels used here, a broad kernel can remove a
 473 winding from that model even though the phase-only equations retain it as
 474 a stationary state.

475 Neither model turns these states into measured cortical spirals. The pe-
 476 riodic sheet cannot carry a lone phase defect, the relaxational dynamics has
 477 no propagation speed, and the amplitude variable has no calibrated relation
 478 to a recorded envelope. Supplemental Material gives the spectral, amplitude
 479 and defect results. Locally, Eq. (5) can still bound content disagreement on a
 480 winding patch when shared encoders are supplied, although its global $r = 0$
 481 form is uninformative. A nonzero ring winding number q also obstructs a
 482 global real-valued phase lift; it has no identified content meaning here. Thus
 483 an onset criterion based only on r can register a change of basin as well
 484 as a change of coupling. The protocol of Section 9.3 needs a spatial phase
 485 observable.

486 8.4 Coherent adaptation can lose environmental align- 487 ment

488 An adaptive coupling rule might supply representational structure, but its
489 objective must reward the relevant agreement. For symmetric coupling, pair-
490 wise sine terms cancel in the population mean drift. Minimizing squared drift
491 at that fixed mean therefore rewards equalizing drift across nodes. In the
492 frequency-cluster control, this favors coupling between unlike-frequency clus-
493 ters, whereas the diagnostic environmental template rewards coupling within
494 clusters.

495 The formalization proves descent of the squared-drift objective for a spec-
496 ified gradient flow at fixed phases. Its functional differs from thermodynamic
497 entropy production: at a coherent stationary density the probability current
498 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
499 rial gives the operator, its gradient and the conditions of the descent result.

500 The finite control co-evolves noisy phases and symmetric, nonnegative
501 couplings at fixed total resource. Its template is a diagnostic covariance
502 of latent signals, never a training input. Across three illustrative seeds,
503 second-half order remains between 0.9167 and 0.9222, while the within-
504 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
505 0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel con-
506 trols are reported in Supplemental Material.

507 The immediate post-update objective reduction disappears when aver-

aged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

8.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on synthetic data separates these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed con-

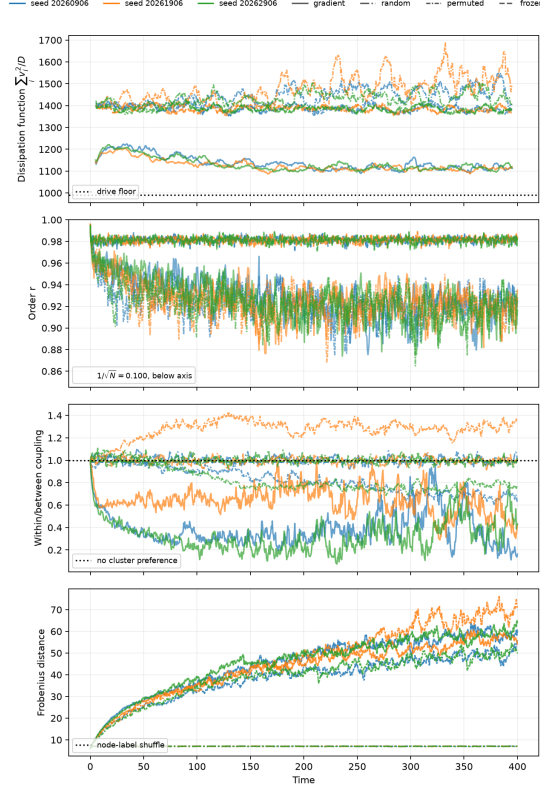


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

530 tent, phase carries a share 1.000 of the content entropy when content is built
531 from it and 0.000 when it is not. A phase-derived statistic reports com-
532 patibility on phase-locked patches whose local descriptions disagree. The
533 implemented content statistic detects their incompatibility. These controls
534 validate a measurement on known constructions; a neural content model and
535 decoder remain to be established.

536 Choosing the cover also chooses a candidate subject: trivial descriptions
537 can agree while carrying little content, and a partition can conceal disagree-
538 ment. The useful choice must therefore be constrained by what neural ac-
539 tivity encodes. The exact composition assumes compatibility of its supplied
540 cover. The content mechanisms above identify additional hypotheses under
541 which agreement can arise; they leave the empirical choice of content and
542 the connection to that cover explicit.

543 **8.6 Estimator calibration on exploratory EEG**

544 The concentration range and the dependence structure decide whether the
545 concentration-coherence relation can be tested at all. Under the specified
546 log-density estimator and bias-floor criterion, discrimination at $a = 0.542$
547 requires 3000 independent sites in the scanned designs, rising to 31000 when
548 sites share a mean in groups of 10. These counts describe that estimator and
549 dependence model at a declared design concentration; the intended recording
550 requires its own calibration. Supplemental Material reports the bin-count
551 controls and the full dependence-aware design.

552 The estimators were also exercised on public propofol-sedation EEG (Ba-
 553 jwa et al., 2025): 61 adjacent bipolar pairs formed from 62 scalp channels,
 554 band-passed, Hilbert-transformed and pooled over pairs and 100-ms win-
 555 dows. Across 8 subjects the pooled concentration spans 0.087–0.142, and
 556 the residual from the Bessel relation is small. That agreement is uninforma-
 557 tive. Pooling over a window does not remove the common rotation of the
 558 phases, so a perfectly phase-locked 10-Hz signal passed through the same
 559 pipeline returns $r = 0.0001$ and $a = 0.0004$ with zero residual, although its
 560 instantaneous spatial resultant is one. The bipolar montage also removes
 561 spatial coherence: a field whose raw channels have instantaneous resultant
 562 0.89 gives 0.06 across bipolar pairs and 0.007 pooled. Any nearly uniform
 563 pooled distribution satisfies $r \approx a/2$, and over the observed range the Bessel
 564 ratio departs from $a/2$ by at most 0.0002. The exercise therefore checks
 565 the estimator’s behaviour and establishes no spatial concentration, coher-
 566 ence level or test of the stationary density. Sedation recordings also leave the
 567 awakening hypothesis untested.

568 An informative dataset therefore needs an estimand that removes the
 569 common rotation, a montage whose effect on spatial phase is modelled, suf-
 570 ficient concentration range or precision, calibrated estimators and an ob-
 571 servation model for dependence and source mixing. Intracranial recordings
 572 (ECoG, sEEG or LFP) or narrowband burst analysis may reach a more
 573 informative concentration range, but that range must be measured. The
 574 design counts above are calculations for the specified estimator, not sam-

575 ple requirements transferable to another recording. Public mouse calcium
576 imaging provides thousands of spatially located neurons across waking, sleep
577 and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the
578 phase-estimation band used here. Site count, state coverage and bandwidth
579 are separate requirements.

580 **9 The cortical hypothesis and an awakening**

581 **test**

582 The cortical proposal identifies a physical source of coupling and asks whether
583 its changes help explain recovery. The tests in Section 8 determine what
584 observations must distinguish before a phase trace can support this identifi-
585 cation.

586 **9.1 Physical identification and calibration**

587 The cortical hypothesis identifies part of effective coupling with endogenous
588 extracellular electromagnetic interactions. Fields can modulate spike timing
589 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
590 propose coordination of distributed neural information (Pinotsis & Miller,
591 2022; McFadden, 2020). Phase organization under propofol motivates study-
592 ing this regime (Bastos et al., 2021), but does not measure coupling.

593 The formal field predicate requires a jointly continuous kernel, a normal-

594 ized substrate, coupling equal to the kernel average, and no single site pro-
595 viding more than half that strength. Toy constructions satisfy it. A cortical
596 realization must specify the scale at which the kernel and its reduction ap-
597 ply: calcium imaging reports different state-dependent modularity patterns
598 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

599 Cortical columns are the obvious candidate unit, and they answer one
600 of the two demands the formalism makes. As the oscillator index they are
601 a serious proposal: they are delineated by response properties and anatomy
602 rather than by the agreement they would be used to establish (Mountcastle,
603 1997), and a phase can be estimated from a column-sized territory at the
604 electrode spacings used for cortical wave measurements (Townsend et al.,
605 2015). As the content cover they are a poor one, for three separate reasons.
606 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
607 the compatibility obligation without constraining anything (Section 4). The
608 cover condition additionally requires the patches to exhaust the substrate,
609 while much of association cortex has no accepted columnar parcellation. And
610 the column’s standing as a canonical cortical unit is contested (Horton &
611 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
612 supplies the oscillator index; the content cover remains an open empirical
613 question, which the bridge assumption E78 records.

614 The spatial reduction is also testable. A row-sum aggregation at zero
615 diagonal tracks the uniform-coupling control over the tested spatial decay
616 range, while frequency heterogeneity displaces the operational threshold. The

617 fit window biases both estimates relative to $2D$; agreement with the control
618 is therefore the supported conclusion. Supplemental Material reports the
619 reduction, its numerical error and the distinct assumptions needed to identify
620 the kernel average with the scalar coupling of the installed-energy condition.

621 The field commitment requires physical evidence beyond a choice of math-
622 ematical representation, and the available support comparisons do not supply
623 it. A finite matrix spread over positive-mass cells preserves its total cou-
624 pling and phase correlation exactly, while point-supported vanishing under
625 an atomless measure and a disadvantage from excluding a better-correlated
626 pair concern specified support constraints; Section 6 reads the same compar-
627 isons against a digital candidate. Neither supplies energy prices.

628 A spike-time shift alone supplies no coupling rate. The dimensionless
629 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
630 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (10)$$

631 This rate describes how induced timing shifts accumulate into drift. Nei-
632 ther it nor the installation factor κ follows from a heat bound. Testing the
633 physical threshold requires independent estimates of the interaction kernel,
634 its aggregation and diffusion in the same cortical regime.

9.2 Extracellular geometry and recovery

The proposed control variable is extracellular geometry. Extracellular volume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density, an effective-medium description predicts that a larger extracellular fraction raises conductivity and reduces field amplitude.

The coupling-gated hypothesis consequently predicts that extracellular contraction on awakening raises E and, holding the other factors in Eq. (10) fixed, raises effective coupling. This argument supplies a direction; the geometry–coupling relation, its magnitude and its role in human recovery require measurement.

For a smooth crossing at nonzero speed, evaluating the stationary branch along the changing control gives a deterministic tracking approximation that fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (11)$$

This is a deterministic tracking approximation from Eq. (3) at fixed D ; the crossing controls in Section 8.2 show that it fails as an observed-onset approximation under the tested initialization. A separation of geometry and phase time scales can support tracking away from threshold. The awakening

655 hypothesis needs a calibrated dynamical prediction for an operational onset
656 above the finite-size fluctuation floor. Crossing K_c alone does not supply a
657 recovery latency.

658 **9.3 A protocol for the awakening experiment**

659 Neural inertia describes a behavioural barrier and hysteresis between anaes-
660 thetic induction and emergence (Friedman et al., 2010); recovery can pass
661 through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
662 2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
663 et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
664 mechanism from a synaptic one. A possible discriminator is spatial onset
665 relative to independently estimated local field coupling and network connec-
666 tivity. The present mean-field model has no spatially varying κ and derives
667 no nucleation order, so a field-based onset map needs a separate model spec-
668 ified before the observations are inspected.

669 For that comparison, partition the recording into prespecified regions i
670 and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score
671 $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements,
672 and a connectivity score c_i from a separately estimated structural network.
673 For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with
674 $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic

675 approximation predicts

$$\widehat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (12)$$

676 Predicted onset is the first crossing of a region-specific threshold fixed from
677 pre-onset fluctuations. Fit the two scales on training sessions, then com-
678 pare held-out onset times and site order under the same observation model.
679 Larger field scores predict earlier onset only when baseline order and diffu-
680 sion do not reverse the ranking. This declared approximation omits interre-
681 gional coupling and saturation; a successful ranking would support its field
682 covariate, not prove ephaptic causation. Supplemental Material specifies the
683 controls and failure conditions.

684 A within-subject protocol can manipulate emergence rate, with spatially
685 resolved recordings to compare the proposed field and connectivity models:

- 686 1. **Manipulate emergence rate.** Vary the propofol infusion taper
687 across sessions within the same subject, recording drug concentrations
688 or a validated pharmacokinetic proxy to quantify the manipulation.
- 689 2. **Measure behavioural recovery.** Record time-to-response to a stan-
690 dard stimulus as the independent behavioural endpoint, rather than
691 estimating ΔK directly.
- 692 3. **Resolve spatial onset.** Identify anatomical locations where coher-
693 ence first crosses the baseline-scaled threshold and score the two maps

694 of Eq. (12) on held-out sessions. The two covariate maps are fixed
695 before the outcome sessions are inspected.

696 **4. Match physical controls.** Confirm that the observed coherence-to-
697 amplitude relation holds under the specific local diffusion D and ob-
698 servation noise. The model’s quadratic-variation rate $2D$ applies to
699 the latent phase; a band-limited analytic phase is smooth below its
700 correlation time, and on a constructed field its squared-increment rate
701 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
702 fore requires an explicit observation model for filtering and sampling.

703 Predeclare the recording window, regional fluctuation thresholds, and a
704 shared timing-error model. Fit the two scales and that observation model
705 on training sessions only; on held-out sessions, score the observed site-onset
706 times and sites that never cross by the same regional predictive rule. Report
707 site order as a secondary outcome and behavioural time-to-response sepa-
708 rately. Sessions without a crossing are censored observations, not zero delays
709 or exclusions. Compare the two scores only where independently estimated
710 field and connectivity maps predict distinguishable onset patterns at the
711 recording resolution. A held-out field-model advantage would support that
712 specified spatial predictor; it would not identify an ephaptic cause, shared
713 content or experience. Supplemental Material gives the controls and pilot
714 quantities needed to judge whether the comparison is informative.

715 10 Discussion

716 10.1 Unity and a minimal self as interpretive motiva- 717 tion

718 We propose the state glued from compatible local descriptions as a formal
719 correlate of conscious unity, and its accurate internal reconstruction as a cor-
720 relate of a minimal self. In the cup example, this would mean one description
721 that preserves the cup’s relation to the perceiver as well as the scene. Com-
722 patibility permits distinct contents and uncertainty; the shared quantities
723 must be chosen from independent evidence about what each region encodes.
724 The reconstruction need not be a verbal or autobiographical account.

725 The minimal self, in Gallagher’s sense, is the pre-reflective sense of be-
726 ing the subject of an experience, present before any narrative or reflective
727 self-description (Gallagher, 2000). Two of its features motivate the proposed
728 correlate. It is not a separate object of experience but a perspective built
729 into the experienced scene; correspondingly, the reconstruction is not a sec-
730 ond representation beside the glued state but a map whose fixed point is that
731 state, read through a part of it. It is immediate rather than inferred from
732 a higher-order state; correspondingly, the fixed point needs no further state
733 that represents the first. The correspondence is structural and does not show
734 that a fixed point is experienced as a perspective. Self-model theory locates
735 the phenomenal self in a transparent model a system has of itself (Met-
736 zinger, 2003), and interoceptive-inference accounts ground it in prediction of

737 the body’s own states (Seth, 2013). The present condition is narrower than
 738 either: it asks only that a declared local encoding recover the global state,
 739 including perceiver-relative variables, at a declared fidelity, and it says noth-
 740 ing about transparency or about which bodily variables must be modelled.
 741 Either account could supply the family of states and the fidelity that the con-
 742 dition leaves as inputs. Under E89 that family is one state at zero error and
 743 a family of diameter at most $2\varepsilon/(1 - \Lambda)$ within error ε (Section 5.2), so the
 744 condition as proved constrains a single self-state; a self that persists through
 745 change needs the input-conditioned map that is not constructed here. Its
 746 proposed relation to experienced perspective remains an interpretation. Nei-
 747 ther identification is among the connecting hypotheses of the composition
 748 (Table 1), and satisfying them would not measure experience. The unity
 749 proposal also lacks an operational definition. The gluing theorem holds for
 750 whatever cover it is given, a disjoint cover satisfies it trivially, and the cover of
 751 interest overlaps in content rather than in space (Section 4); until a principle
 752 fixes that cover, the correlate is a schema awaiting one.

753 **10.2 A comparison that could disconfirm the link to** 754 **content**

755 The framework predicts that conscious content tracks compatibility, not co-
 756 herence. The comparison must also separate compatibility from how well
 757 each region encodes the stimulus, because two accurate decoders agree auto-

758 matically. If decoders d_A and d_B estimate a shared stimulus quantity x with
759 errors $e_A = d_A - x$ and $e_B = d_B - x$, then

$$|d_A - d_B| \leq |e_A| + |e_B|, \quad |d_A - d_B|^2 = |e_A|^2 + |e_B|^2 - 2\langle e_A, e_B \rangle. \quad (13)$$

760 Consciously perceived stimuli are usually decoded better, so higher agreement
761 on those trials would follow from single-region fidelity alone. At matched
762 single-region error, agreement differs only through the covariance of the two
763 errors, which is the term compatibility bears on: two regions describing one
764 content should err together.

765 Present matched stimuli that are perceived on some trials and not on
766 others, for example at masking or rivalry threshold, and fit decoders for
767 one shared stimulus quantity independently on two regions that both encode
768 it. Classify trials by perceptual state with a no-report measure, such as
769 reflexive eye movements or pupil responses that track the dominant percept
770 in rivalry (Tsuchiya et al., 2015), so that the classification does not also
771 select for the neural correlates of reporting; report-based classification is a
772 secondary analysis. Match the two classes on phase coherence between the
773 regions and on each region’s own decoding error (or, where the stimulus
774 value is uncertain, on each region’s signal strength). The statistic is the
775 excess agreement over a fidelity-matched null: pair region A ’s decode on one
776 trial with region B ’s decode on another trial of the same stimulus and error
777 stratum, which preserves both marginal accuracies and removes any coupling

778 of the errors.

779 The proposed link predicts a larger excess on perceived trials at matched
780 coherence and matched single-region accuracy (Fig. 6). Equal excess across
781 classes, or excess that follows coherence and not perceptual state, counts
782 against it. A positive result is not yet specific: noise shared upstream of both
783 regions also correlates their errors, so the excess should be compared against
784 an upstream region’s decoding error, matched in the same way. A self-model
785 comparison has the same form: at matched accuracy for the scene variable,
786 the reconstruction of perceiver-relative variables, such as hand position rel-
787 ative to the target, from a declared local encoding should be more accurate
788 on perceived trials. These are correlational tests of proposed correlates, and
789 the no-report classification is itself an indirect marker of experience.

790 **10.3 What a test of the full proposal must establish**

791 A test needs separate estimates of effective coupling and diffusion, shared
792 content on overlapping regions, and reconstruction from a specified encoding
793 at a declared fidelity. The cover, relevant state family and decoder must
794 be fixed without selecting them for agreement. An independent experiential
795 endpoint is needed to evaluate the interpretation; absence of a report alone
796 does not define absence of experience. A unified episode with incompati-
797 ble decoded descriptions would challenge the unity identification. A unified
798 episode lacking the proposed self-encoding would challenge the minimal-self
799 claim. Conversely, satisfying the mathematical conditions without such an

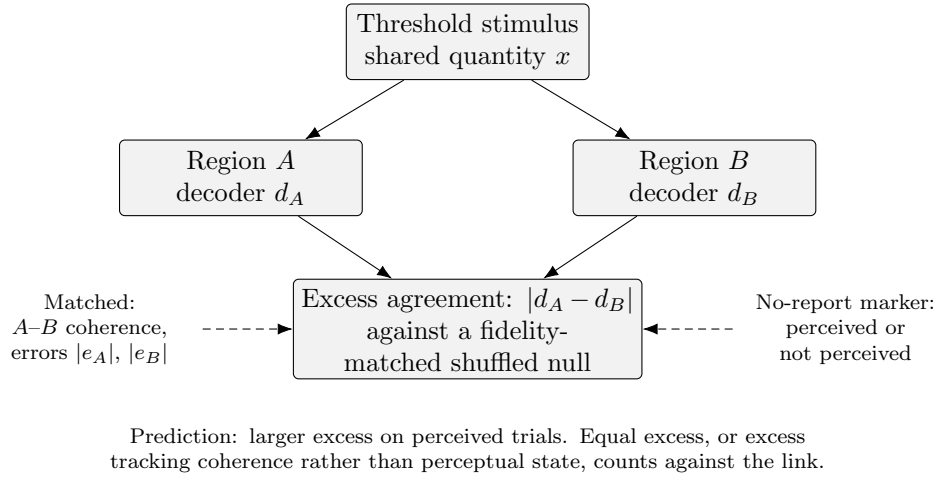


Figure 6: Design of the disconfirming comparison. Decoders for one shared stimulus quantity are fitted independently on two regions. Trials are classed by a no-report marker of perceptual state and matched on inter-regional phase coherence and on each region’s own decoding error, so that agreement cannot follow from single-region fidelity (Eq. (13)). The statistic is agreement in excess of a null that pairs decodes across trials within the same stimulus and error stratum.

800 episode would challenge their sufficiency as correlates.

801 For the physical route, cortical actuator modes and prices, the geometry–
 802 coupling relation and a dynamical onset model remain to be measured. The
 803 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
 804 diction. Controlled changes in coordination and independently decoded con-
 805 tent agreement would address the representational connection. The synthetic
 806 decoder exercise calibrates part of these measurements, and the EEG exercise
 807 identifies what a pooled bipolar estimand cannot measure; neither identifies
 808 a neural content code or tests experience.

809 10.4 Explanatory scope

810 Integrated-information, global-workspace and predictive-processing accounts
811 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
812 1997; Dehaene et al., 2014; Friston, 2010). This account specifies conditional
813 relations among coupling, local content and reconstruction. Its formal field
814 condition requires a continuous kernel, a normalized substrate, a specified
815 average coupling and noise, and a site-contribution bound. Those conditions
816 do not identify an electromagnetic substrate; the extracellular interpretation
817 needs the separate calibration of Section 9. The joint witness shows that
818 the formal hypotheses can hold together in a declared model. A cortical
819 realization and an experiential interpretation require additional evidence.

820 Four limitations bound every claim above. Synchrony is not itself an
821 experiential measure, and nothing here establishes a consciousness criterion:
822 the conditions are proposed correlates to be tested against report or another
823 independent endpoint. The present EEG analysis checks a measurement
824 pipeline and does not validate any experiential claim. The formal results are
825 conditional on declared encoders, covers and maps whose cortical counter-
826 parts are unmeasured. No principle selects the cover on which compatibility
827 is assessed, so the unity correlate is not yet operationally defined.

828 The remaining explanatory question is whether these relations occur in
829 neural activity and matter for experience. The framework identifies candidate
830 correlates and tests of their physical and representational premises. It does
831 not explain why satisfying them would be accompanied by experience.

832 11 Conclusions

833 Phase coherence, compatibility and self-reconstruction are separable condi-
834 tions. Coherence constrains shared content only through encoders shared on
835 overlaps and with a bound that grows as the square root of the population
836 read. That bound is vacuous beyond 10 sites at the simulated sheet’s patch
837 order, so for populations of cortical size the positive link from coherence
838 to content is the bridge assumption E78 and not a derived result. Com-
839 patible local descriptions, not synchrony, determine one global state; and a
840 contracting self-map fixes that state while reconstructing only a family of
841 bounded diameter. Each result replaces an assumption about synchrony by
842 a measurement: independently decoded agreement on overlaps, a declared
843 encoding and its fidelity, and calibrated coupling and diffusion. The compar-
844 ison of Section 10.2 states how such measurements could count against the
845 proposed link to conscious content.

846 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on

Table 2 (continued)

Symbol	Defined	Meaning
D	Eq. (1), Supp.	Phase diffusion in inverse time; in the Supplemental Material's convergence proof, the phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
U_{inst}, κ	Section 3.2	Installed energy of a declared actuator and the mode-data factor bounding its mean coupling by κU_{inst}
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (9)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (7)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time; in Eq. (4), τ is the duration of a change
J, σ_J, C	Eq. (4)	Probability current of the phase density, its current functional and a current budget
T	Section 5	Number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from
$\gamma, \mathcal{T}, f$	Eq. (10)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph

Table 2 (continued)

Symbol	Defined	Meaning
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover’s nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (9)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (7)	Encoding region and the reconstruction map $F(s) = \text{readout}(s _{\mathcal{A}})$
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (10)	Field amplitude, in the cortical identification
s, s_*	§4–5	A global section and the one the reconstruction fixes

847 Data and code availability

848 The Lean development, simulation scripts, tests and saved numerical sum-
849 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
850 `cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Math-
851 lib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publication
852 simulation values are generated from saved summaries without rerunning
853 production sweeps. The exploratory EEG analysis uses OpenNeuro dataset
854 `ds005620` (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045,
855 1054, 1060, 1067, resting awake and sedation runs 1–2); preprocessing
856 and selection are described in Supplemental Material and implemented in

857 `simulations/empirical_collapse.py`. The repository contains the repro-
858 ducible analysis and does not redistribute the raw recordings.

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862 **Use of AI tools**

863 AI assistance to this work was substantial and is not separable by task.
864 Several coding assistants and chat interfaces were used in an interleaved way
865 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
866 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-
867 5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),
868 Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek
869 v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them
870 they contributed to the Lean development, the simulation and analysis code,
871 and the drafting and revision of both documents. Because the tools were
872 used in combination and in alternation, no section, proof, figure or result is
873 attributable to any one of them, and none is claimed to be. The author is
874 responsible for all content, including everything a tool produced. Machine
875 checking establishes the stated Lean results under their hypotheses; it does

876 not validate biological interpretations or replace scientific review of generated
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